F12 7th MRL

MAC 1147 Review 1, Fall 2012
Exam 1 covers Lectures 1-10

1b. Write without absolute value signs and simplify:

$|\pi - 5| - |3 - \pi|$

neg neg

$-(\pi-5) - -(3-\pi) =$

$-(\pi-5) + (3-\pi) =$

$-\pi + 5 + 3 - \pi = \boxed{8 - 2\pi}$

4e. Factor completely:

$$2x(4-3x)^{4/3} - 4x^2(4-3x)^{1/3}$$

$$2x(4-3x)^{1/3}\left[(4-3x)^{4/3-\frac{1}{3}} - 2x\right] =$$

$$2x(4-3x)^{1/3}\left[(4-3x)^{1} - 2x\right] =$$

$$\boxed{2x(4-3x)^{1/3}(4-5x)}$$

lcd: $x(x+1)$

$1-\frac{1}{x} \neq 0$

$1 = \frac{1}{x}$ $\quad x \neq 1$

7b. Simplify each complex fraction. State the domain of fraction in (b).

D: $x \neq 0$
$x \neq -1$
$x \neq 1$

$$\left(\frac{\frac{3}{x} - \frac{4}{x+1}}{1 - \frac{1}{x}}\right)\left(\frac{x(x+1)}{x(x+1)}\right) = \frac{3(x+1) - 4x}{x(x+1) - (x+1)} =$$

$$= \frac{3x+3-4x}{x^2+x-x-1} = \boxed{\frac{-x+3}{x^2-1}} = \boxed{\frac{3-x}{x^2-1}, \; x \neq 0}$$

implied $x \neq \pm 1$

7c. $\dfrac{x^3(1-x)^{-1/2} + x(1-x)^{1/2}}{x^2} =$

$$\frac{x(1-x)^{-\frac{1}{2}}\left[x^2 + (1-x)^{\frac{1}{2}-(-\frac{1}{2})}\right]}{x^2} = \frac{x(1-x)^{-\frac{1}{2}}\left[x^2+(1-x)\right]}{x^2}$$

$$= \frac{x(x^2+1-x)}{x^2(1-x)^{1/2}} = \boxed{\frac{x(x^2-x+1)}{x^2(1-x)^{1/2}}} = \boxed{\frac{x^2-x+1}{x(1-x)^{1/2}}}$$

8a. Simplify each expression. Write answer using only positive exponents.

$$\frac{27(-3xy^{-2})^{-3}}{x^2y^4} = \frac{27(-3)^{-3}x^{-3}y^{6}}{x^2y^4} =$$

$$= \frac{27\ y^6}{(-3)^3x^3x^2y^4} = \frac{\cancel{27}y^6}{-\cancel{27}x^5y^4} = \boxed{-\frac{y^2}{x^5}, y \neq 0}$$

9a. Simplify and rationalize the denominator where it is necessary:

$$\sqrt[5]{\frac{-\cancel{4}x^{10}y^{12}}{\cancel{64}x^{-5}y^{19}z^3}} = \sqrt[5]{\frac{-x^{15}}{16y^7z^3}} = \frac{\sqrt[5]{-x^{15}}}{\sqrt[5]{16y^7z^3}} =$$

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$$= \frac{\sqrt[5]{-x^{15}}}{\sqrt[5]{16y^7z^3}}\left(\frac{\sqrt[5]{2y^3z^2}}{\sqrt[5]{2y^3z^2}}\right) = \frac{\sqrt[5]{-x^{15}\cdot 2\cdot y^3\cdot z^2}}{\sqrt[5]{2^5y^{10}z^5}} =$$

2^4

$$= \frac{\sqrt[5]{-1(x^3)^5(2)y^3z^2}}{\sqrt[5]{2^5(y^2)^5z^5}} = \frac{-1x^3\sqrt[5]{2y^3z^2}}{2y^2z}$$

$$= \boxed{-\frac{x^3\sqrt[5]{2y^3z^2}}{2y^2z}}$$

$\sqrt[5]{-1} = -1$

12c. Simplify each radical expression if possible:

D:

$x \geq 0$

$x \neq 0$

$x \neq 1$

$$\frac{\left(\frac{1}{\sqrt{x}} - \sqrt{x}\right)^2}{(x-1)^2} = \frac{\frac{1}{x} - 2\left(\frac{1}{\sqrt{x}}\right)(\sqrt{x}) + x}{(x-1)^2} =$$

$$= \left(\frac{\frac{1}{x} - 2 + x}{(x-1)^2}\right)\left(\frac{x}{x}\right) = \frac{1 - 2x + x^2}{x(x-1)^2} = \frac{x^2 - 2x + 1}{x(x-1)^2} =$$

$$= \frac{\cancel{(x-1)^2}}{x\cancel{(x-1)^2}} = \boxed{\frac{1}{x},\ x > 0 \text{ and } x \neq 1}$$

17c. Solve the equations:

$$\sqrt{5x+4} - 2\sqrt{x} = 1$$

$$\left(\sqrt{5x+4}\right)^2 = \left(2\sqrt{x}+1\right)^2$$

$$5x+4 = 4x + 2(2\sqrt{x})(1) + 1$$

$$5x+4 = 4x + 4\sqrt{x} + 1$$

$$(x+3)^2 = (4\sqrt{x})^2$$

$$x^2+6x+9 = 16x$$

$$x^2-10x+9=0$$

$$(x-9)(x-1)=0$$

$$\boxed{x=9 \qquad x=1}$$ must check sol.

$x=9$

$$7-6=1$$
$$1=1 \checkmark$$

$x=1$

$$\sqrt{9}-2(1)=1$$
$$3-2=1$$
$$1=1 \checkmark$$

19d. Solve the absolute value equations:

$$|x^2 - 2x| = 8$$

case 1: If x^2-2x is pos.

$x^2-2x=8$

$x^2-2x-8=0$

$(x-4)(x+2)=0$

$\boxed{x=4 \quad x=-2}$ Must check.

$|4^2-2(4)|=8$

$|16-8|$

$8=8$ ✓

$|(-2)^2-2(-2)|=8$

$|4+4|=8$

$8=8$ ✓

$-(x^2-2x)=8$

$-x^2+2x=8$

$(-1)(-x^2+2x-8)=0(-1)$

$x^2-2x+8=0$

case 2: If x^2-2x is neg

$(+1)[+(x^2-2x)]=[8](+1)$

$x^2-2x=-8$

$x^2-2x+8=0$

$x=\frac{2\pm\sqrt{4-4(8)}}{2}$

$x=\frac{2\pm\sqrt{-28}}{2}$

no real solutions

20d. Solve the absolute value inequalities. Give answer in interval notation.

$$3 - |x+1| < \frac{1}{2}$$

$(-1)(-|x+1|) < (-\frac{5}{2})(-1)$ mult bothsides by -1

$|x+1| > \frac{5}{2}$

case 1: If $x+1$ is pos

$x+1 > \frac{5}{2}$

$x > \frac{3}{2}$

or

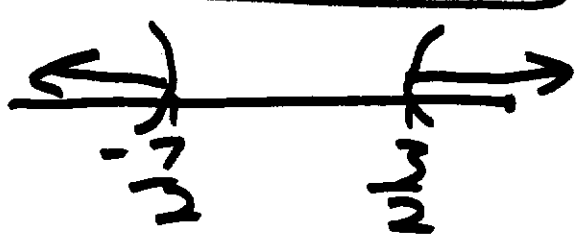
case 2: If $x+1$ is neg.

$(-1)(-(x+1)) > \frac{5}{2}(-1)$

$x+1 < -\frac{5}{2}$

$x < -\frac{7}{2}$

$\boxed{(-\infty, -\frac{7}{2}) \cup (\frac{3}{2}, \infty)}$



$\left(\frac{1}{2}\cdot 4\right)^2 = (2)^2 = 4$

$\left(\frac{1}{2}(-2)\right)^2 = (-1)^2$
$= 1$

26. Find the center and radius of the circle.

$\frac{1}{3}\left(3x^2 + 3y^2 + 12x - 6y - 1 = 0.\right)$

$x^2 + y^2 + 4x - 2y - \frac{1}{3} = 0$

$\underbrace{x^2 + 4x + 4} + y^2 - 2y + 1 = \frac{1}{3} + 4 + 1$

$(x+2)^2 + (y-1)^2 = \frac{16}{3}$

$C(-2, 1)$

$r^2 = \frac{16}{3} \rightarrow r = \sqrt{\frac{16}{3}} = \frac{4}{\sqrt{3}}$

$(x-h)^2 + (y-k)^2 = r^2$

$r = \frac{4\sqrt{3}}{3}$

x	y
0	0

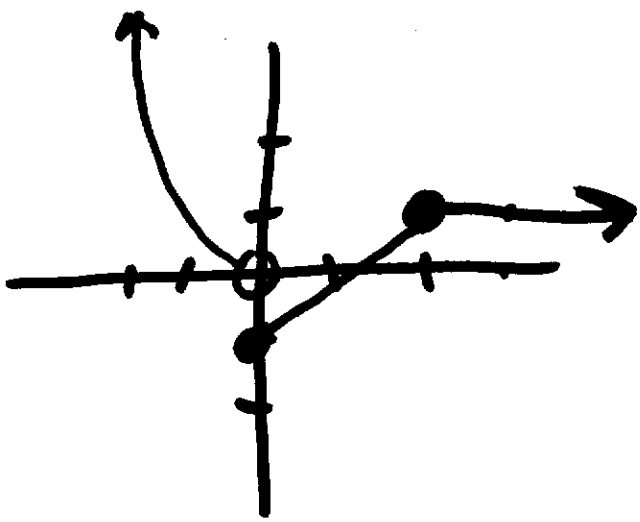
30. Given the piecewise defined function:

$$f(x) = \begin{cases} x^2 & \text{if } x < 0 \\ x - 1 & \text{if } 0 \le x < 2 \\ 1 & \text{if } x \ge 2 \end{cases}$$

x	y
0	-1
2	1

x	y
2	1
3	1

Find $f(-2)$, $f(0)$, $f(1)$, $f(2)$ and $f(4)$. On what open intervals is the function increasing, decreasing and constant? Find the local maximum and minimum values of the function if they exist.



36. List all reflections, translations and stretching that are needed in order to obtain the graph of $f(x) = -2\sqrt{-x} + 1$ from the graph of $y = \sqrt{x}$. Find the domain of f and sketch its graph. What is the range of $f(x)$?